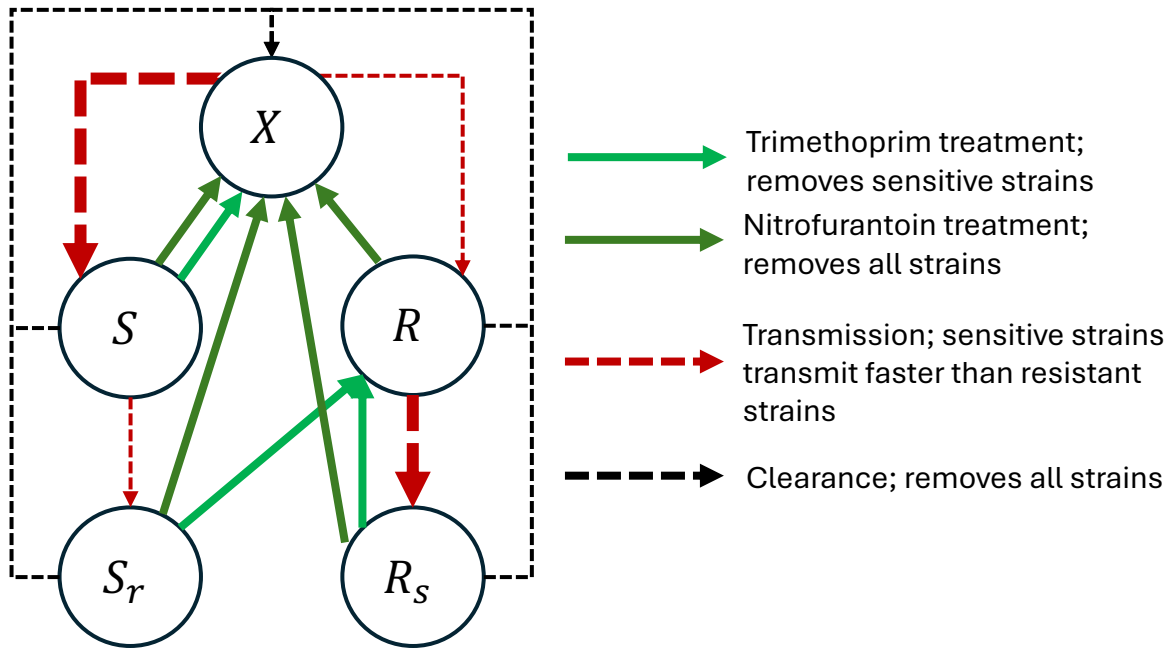




$$X = 1 - (S + R + S_r + R_s)$$

$$\lambda_S = \beta(S + S_r)$$

$$\lambda_R = (1 - c)\beta(R + R_s)$$

$$\tau_T(t) = p_{UTI} \cdot p_{treat} \cdot p_{trim}(t) \cdot \tau$$

$$\tau_N(t) = p_{UTI} \cdot p_{treat} \cdot (1 - p_{trim}(t)) \cdot \tau$$

$$\frac{dS}{dt} = \lambda_S X - \lambda_R S - \mu S - (\tau_T(t) + \tau_N(t))S$$

$$\frac{dR}{dt} = \lambda_R X - \lambda_S R + \tau_T(t)(S_r + R_s) - \mu R - \tau_N(t)R$$

$$\frac{dS_r}{dt} = \lambda_R S - \mu S_r - (\tau_T(t) + \tau_N(t))S_r$$

$$\frac{dR_s}{dt} = \lambda_S R - \mu R_s - (\tau_T(t) + \tau_N(t))R_s$$